

Q1. Problem

A network consists of the following topology:

PC-A — SW1 — R1 — R2 — SW2 — PC-B

Given:

Packet Size = **1,500 bytes**

Links

PC-A ↔ SW1: **100 Mbps**, propagation = **1 μs**

SW1 ↔ R1: **100 Mbps**, propagation = **2 μs**

R1 ↔ R2: **10 Mbps**, propagation = **10 ms**

R2 ↔ SW2: **100 Mbps**, propagation = **2 μs**

SW2 ↔ PC-B: **100 Mbps**, propagation = **1 μs**

Assumptions

- Store-and-forward at switches and routers
- No queuing delay
- Ignore processing delay
- No fragmentation
- Full-duplex links

Questions

(a) Compute end-to-end delay

1500 bytes = 1500 × 8 = 12000 bits

$d_{trans} =$

1. **100 Mbps links** $\Rightarrow 12000 / (100 \times 10^6) = 120 \times 10^{-6} \text{ s}$

There are **4 such links**:

- PC-A → SW1
- SW1 → R1
- R2 → SW2
- SW2 → PC-B

$4 \times 120 \mu\text{s} = 480 \mu\text{s} = 0.48 \text{ ms}$

2. **10 Mbps link** (R1 ↔ R2):

$12000 / (10 \times 10^6) = 1.2 \text{ ms}$

3. Propagation delays

$1 + 2 + 10,000 + 2 + 1 \mu\text{s} = 10,006 \text{ ms}$

4. Total delay

$d_{total} = d_{transmission} + d_{propagation} = (0.48 + 1.2) + 10.006 = 1.68 + 10.006$

11.686 ms

(b) Identify the bottleneck link

Compare transmission rates:

- 100 Mbps
- **10 Mbps** ← smallest **R1 $\leftrightarrow$ R2**

(c) After first packet, will switches flood or unicast? Explain briefly.

First packet:

- Switches don't know MACs → **flood**

After learning:

- MAC tables populated → **unicast forwarding**

(d) If PC-A sends 10 packets back-to-back, what is the effective throughput?

Pipeline forms → throughput limited by **slowest link**

That's the **10 Mbps link**

So once pipeline is full: Throughput=10 Mbps

Time per packet on bottleneck: 1.2 ms

For 10 packets: $10 \times 1.2 = 12$ ms

Q2. Bandwidth-Delay Product

Link:

- Rate = 50 Mbps
- Propagation delay = 20 ms

(a) Bandwidth-delay product

$50 \times 10^6 \times 0.02 = 1,000,000$ bits

(b) How many packets (1,000 bytes)?

$1,000,000 / (8000) = 125$ packets

Q3. Fragmentation

An IP packet:

- Size = 4000 bytes
- MTU = 1500 bytes
- Header = 20 bytes

Payload per fragment:

$1500 - 20 = 1480$ bytes

Number of fragments:

$4000 / 1480 \approx 2.7 \Rightarrow 3$ fragments

Fragment sizes:

- $1480 + \text{header} \rightarrow 1500$
- $1480 + \text{header} \rightarrow 1500$
- Remaining:

$4000 - 2960 = 1040 + \text{header} = 1060$

3 fragments

Sizes: **1500, 1500, 1060** bytes

Q4. Multihop bottleneck

Path:

- 100 Mbps
- 50 Mbps
- **20 Mbps**
- 80 Mbps